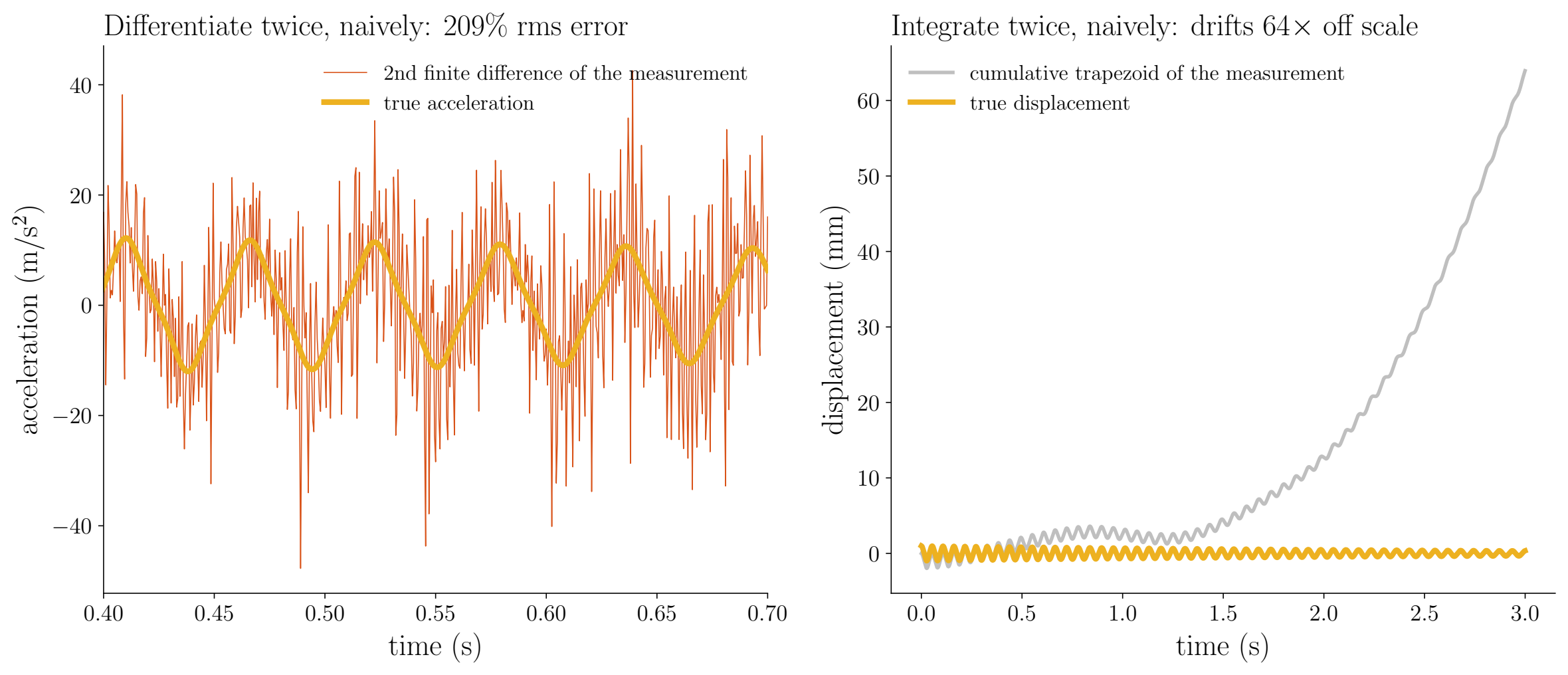

Differentiate twice, naively: 209% rms error
2nd finite difference of the measurement
true acceleration
acceleration (m/s^2)
40
20
0
−20
−40
0.40
0.45
0.50
0.55
0.60
0.65
0.70
time (s)
Integrate twice, naively: drifts 64× off scale
cumulative trapezoid of the measurement
true displacement
displacement (mm)
60
50
40
30
20
10
0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
time (s)